CENG381 Stochastic Processes

CTMC and Birth-Death Processes — Practice 1

Q1 (30 points). A machine operates in three states: 0 = Idle, 1 = Working, 2 = Broken. The continuous-time transition rates are:

- Idle → Working at rate 2 per hour
- Working → Idle at rate 3 per hour
- Working → Broken at rate 1 per hour
- Broken → Working at rate 2 per hour

a) Write the 3×3 rate matrix Q. Verify that each row sums to 0.

b) Find the stationary distribution π by solving $\pi \cdot Q = 0$ and $\pi_0 + \pi_1 + \pi_2 = 1$.

c) Interpret the result: what fraction of time is the machine idle, working, and broken? What is the expected return time to the Working state?

Q2 (35 points). A hospital ward has at most 3 patients at any time. The number of patients follows a birth-death process on {0,1,2,3}:

- Birth rates (new admissions): $\lambda_0=4$, $\lambda_1=3$, $\lambda_2=2$
- Death rates (discharges): $\mu_1=2$, $\mu_2=3$, $\mu_3=4$

a) Write the 4×4 rate matrix Q for this birth-death process.

b) Use the detailed-balance equations $\lambda_n \pi_n = \mu_{n+1} \pi_{n+1}$ to express π_1, π_2, π_3 in terms of π_0. Then normalise to find all four probabilities.

c) Compute the expected number of patients $E[N] = \Sigma\, n \cdot \pi_n$. Also find P(ward is full) = π_3.

Q3 (35 points). A frog lives near a pond. It jumps from land (state 1) to water (state 0) at rate $\alpha = 3$ per hour, and from water to land at rate $\beta = 1$ per hour.

a) Write the rate matrix Q. Find the stationary distribution π. Interpret: what fraction of time does the frog spend in water?

b) Construct the embedded discrete-time chain P by uniformising at rate $r = \max(q(0), q(1)) = 3$. Verify that P has the same stationary distribution as the CTMC.

c) The state probabilities evolve as $P_1(t) = \pi_1 + C \cdot e^{\lambda_2 t}$, where λ_2 is the non-zero eigenvalue of Q. If the frog starts on land ($P_1(0) = 1$), find C and write the explicit formula for $P_1(t)$. How long does it take for $|P_1(t) - \pi_1|$ to fall below 0.05?